Quarto Workshop

From data to formatted publication

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Quarto lets us…

- Analyze data and visualize data through R and Python (and more): **You know - the things we do anyway**
- Create dynamic content for scientific reports, manuscripts or presentations: **Create output when input changes**
- Using one source and produce output in various formats like HTML, PDF and MS Word: **Reduce copy-paste**
- Create reports, manuscripts, presentations, websites, books or knowledge repos through Pandoc and writing in Markdown
- Do all Open-Source and adopt good Open-Science practice

> Provide seamless and reproducible workflows from data analysis to dissemination of knowledge

Aims of this workshop

- After this workshop you can understand and describe the basic mechanisms of Quarto and know where it can support your research activities.
- Apply Quarto to concrete use-cases to get inspiration about its capabilities.
- You will have a running toolchain on your computer to work further on your own projects.

What we don’t aim

- Introduce coding basics, R programming or Markdown editing.
- Workshops to R and ggplot were given in the past and resources are available in this Gitlab repository
- Quarto is huge. We won’t cover everything.

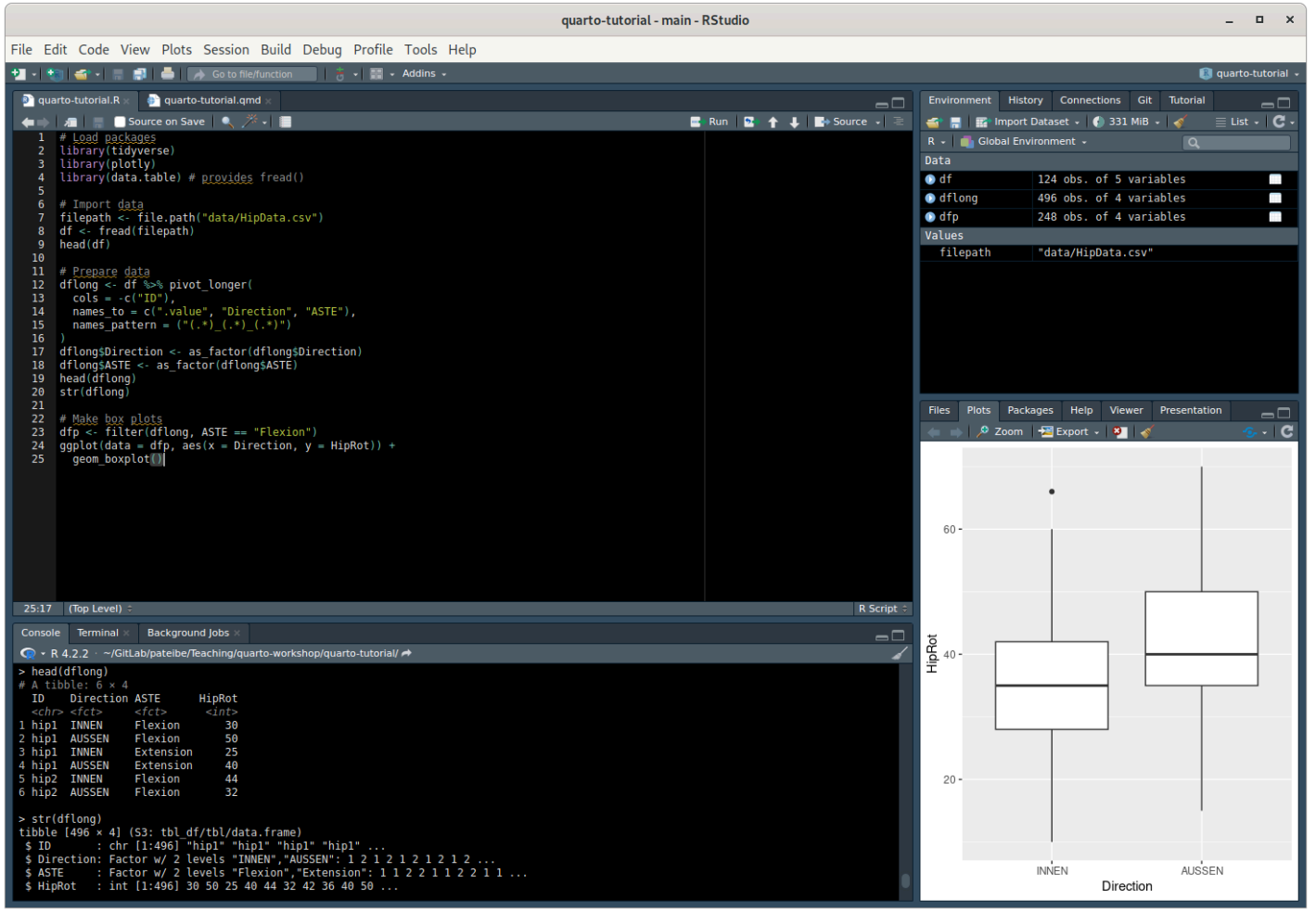
Helpful resources and tips

- Quarto website. Lot of material > good to take time to familiarize yourself. The Guide section is a good starting point.
- Adopt using Visual Studio Code, a powerful text editor.
- Check out the getting started tutorials for VS Code or RStudio on the Quarto website.
- Familiarize with basic Markdown authoring.
- RStudio approach is easier to start for R users.
- VS Code workflow is more versatile but harder to adopt.

Workshop tools

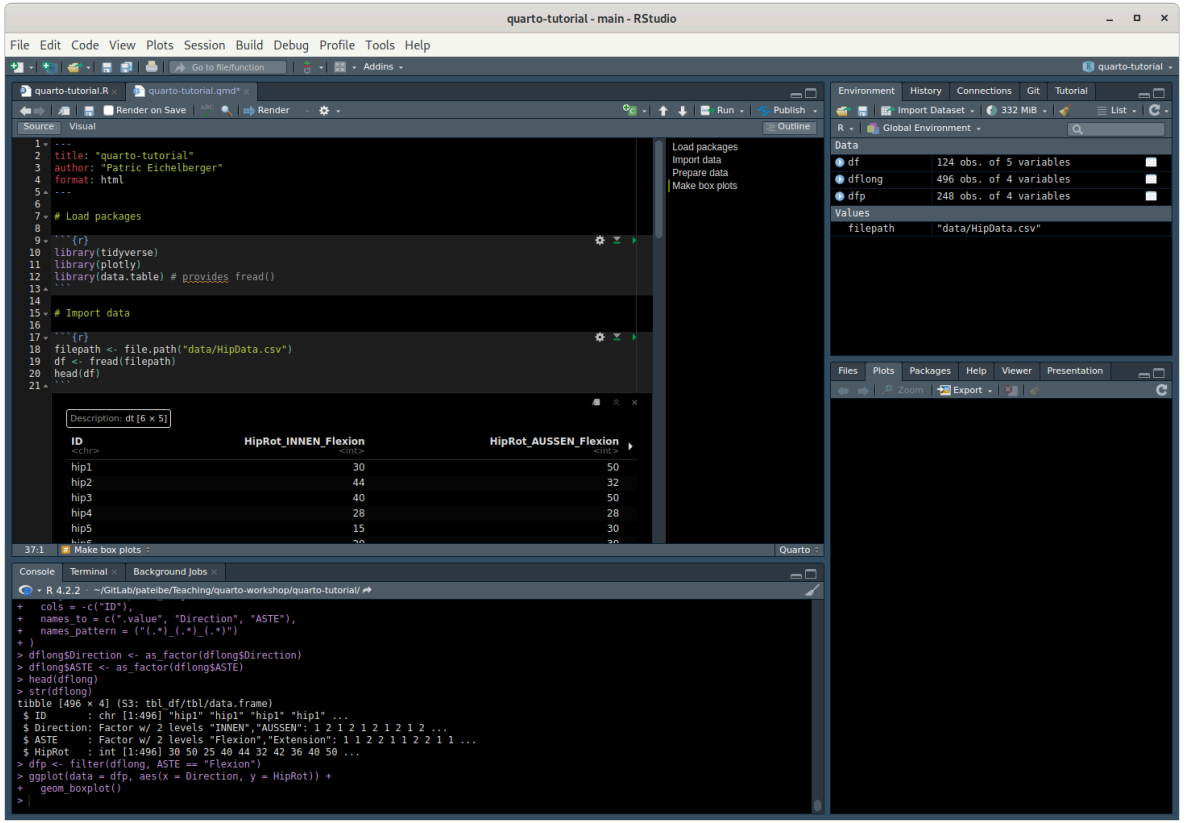
- R
- RStudio
- Visual Studio Code
- Quarto

R-Script (.R)



Outputs go to the console and to the plot pane.

Quarto document (.qmd)



- Outputs are inline within the `.qmd` file when run (Button **Run**) in RStudio.
- Text, code and output go to `.html` (or whatever format we set in the YAML-header) when rendered through Quarto (Button **Render**).

How it works?

qmd file with R code in RStudio

qmd file with Python code from VS Code (or command line)

YAML header

Where we set rendering options

```yaml
---
title: "quarto-tutorial"
author: "Patric Eichelberger"
format: html
---
```

- Common and format-specific rendering options.
- Multiple formats can be specified to produce different output from the same source.

More complex

```yaml
---
toc: true
bibliography: references.bib
lang: de
date: today
csl: ieee.csl
format:
  pdf:
    documentclass: scrreprt
    papersize: a4
    number-sections: true
    margin-left: 30mm
    margin-right: 30mm
    mainfont: Unit Rounded Pro
    mainfontoptions:
      - UprightFont = *-Light
  html:
    toc: true
    embed-resources: true
---
```

Learn more: YAML html options

Computations

- Computations are carried out within code chunks.
- Execution options can be set with `#|`.

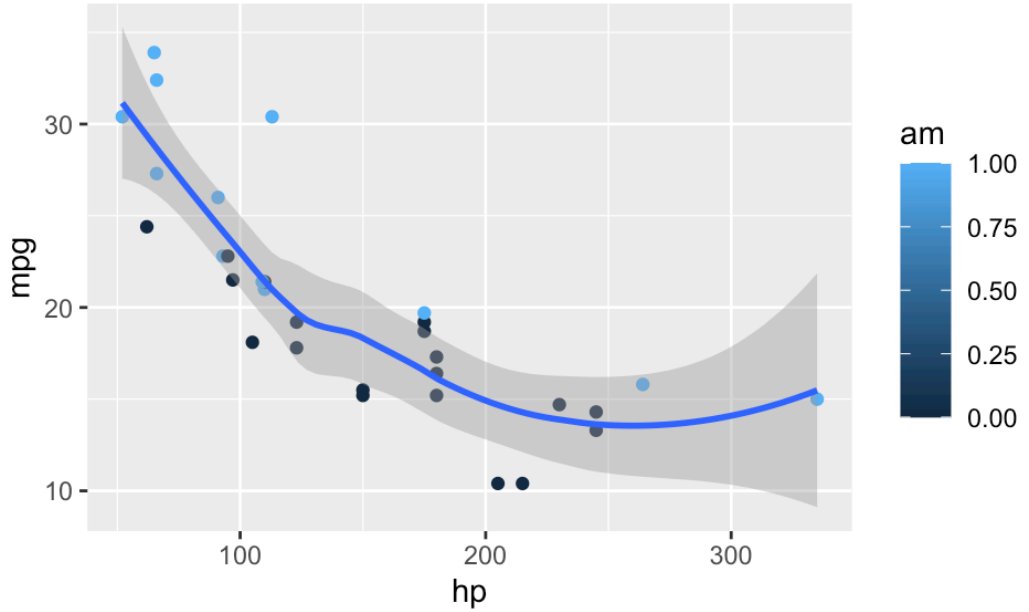
````
```{r}
#| label: fig-hiprotation # the chunk can be referred to by its label
#| fig-cap: "Hip internal and external rotation." # caption
#| eval: true # evaluate chunk when set to true
#| include: true # include output when set to true
#| warning: false # include warnings in the output
#| fig-width: 5 # set figure width to 5 inches

print("Hello World!")
```
````

```
[1] "Hello World!"
```

Learn more: Execution options

Example dynamic content

```
1 library(ggplot2)
2 ggplot(mtcars, aes(hp, mpg, color = am)) +
3   geom_point() +
4   geom_smooth(formula = y ~ x, method = "loess")
```



The plot is directly created during rendering and not imported from an external file.